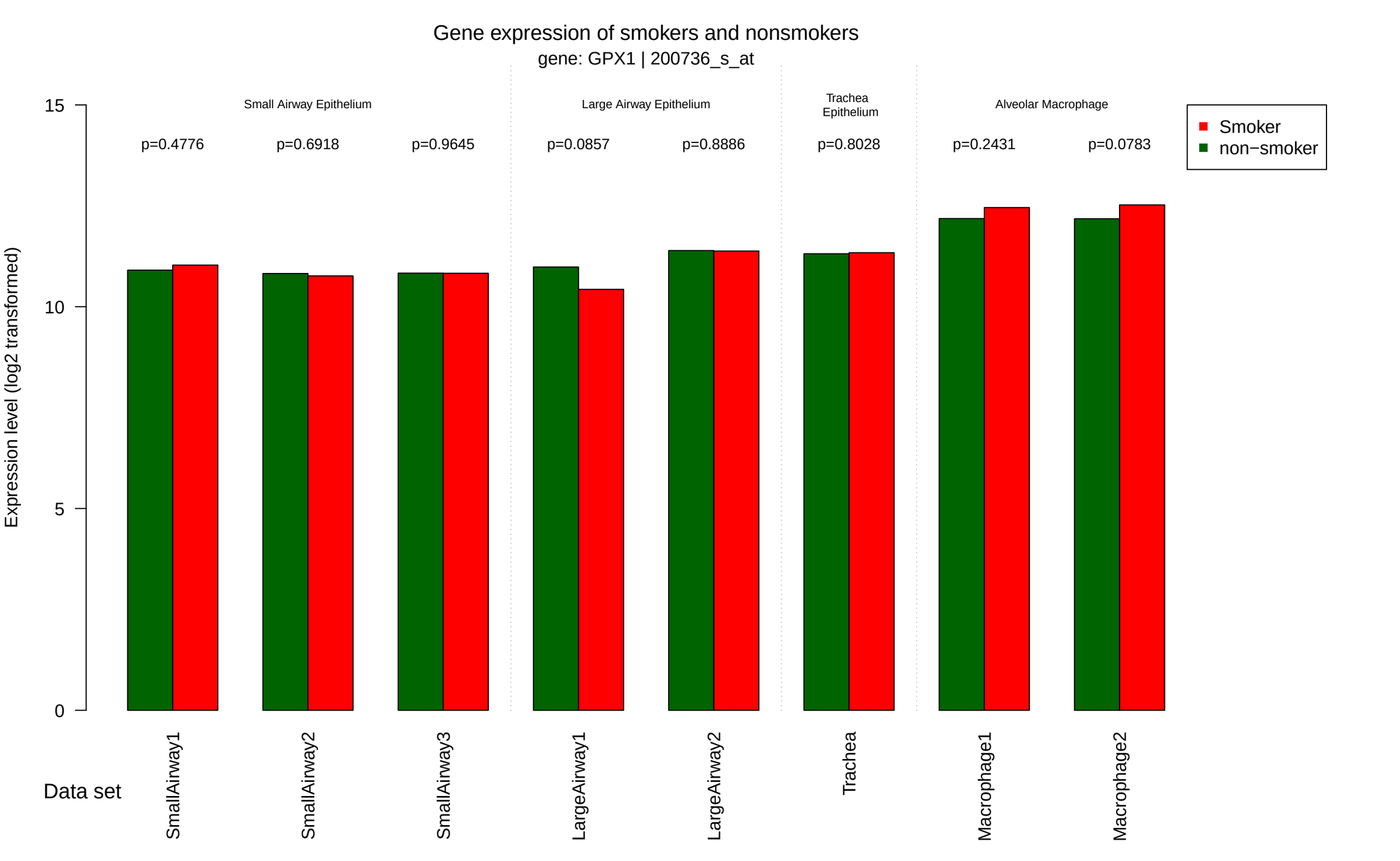
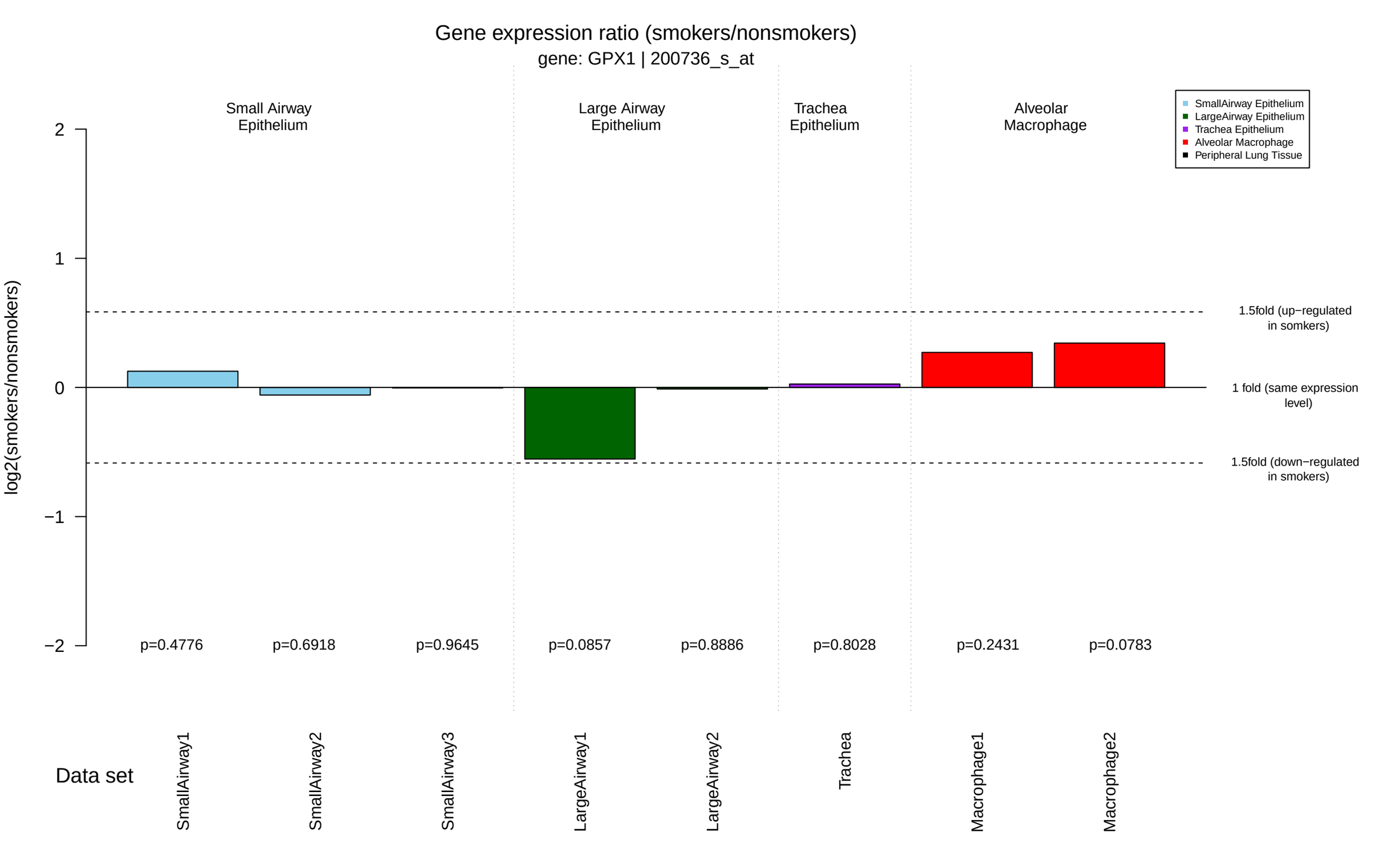
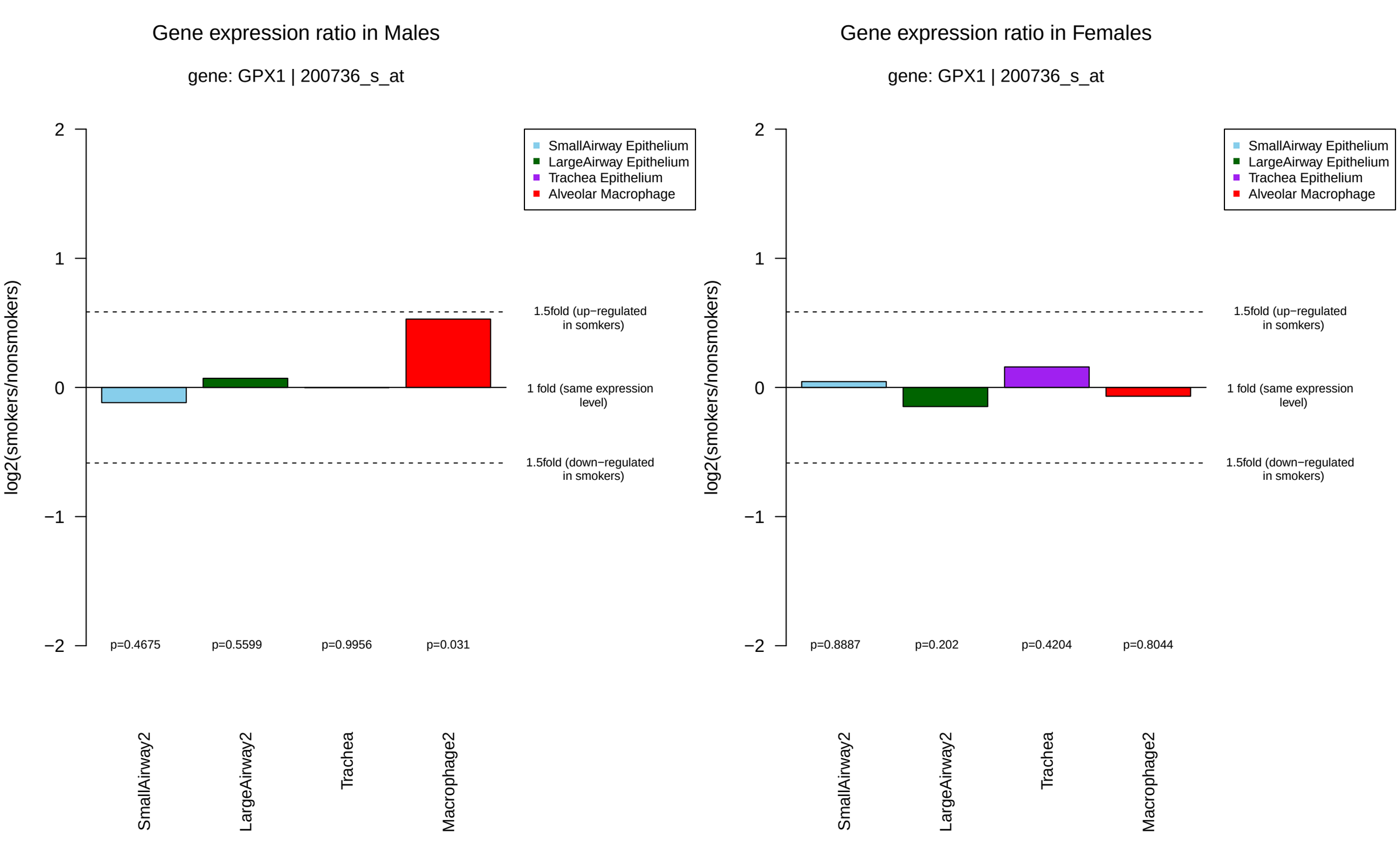
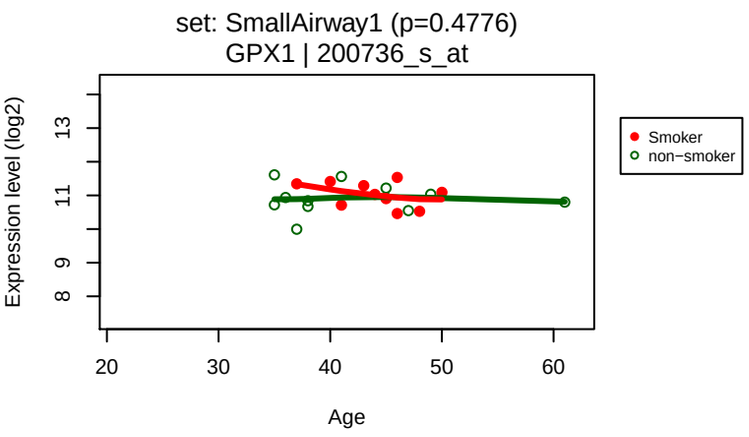
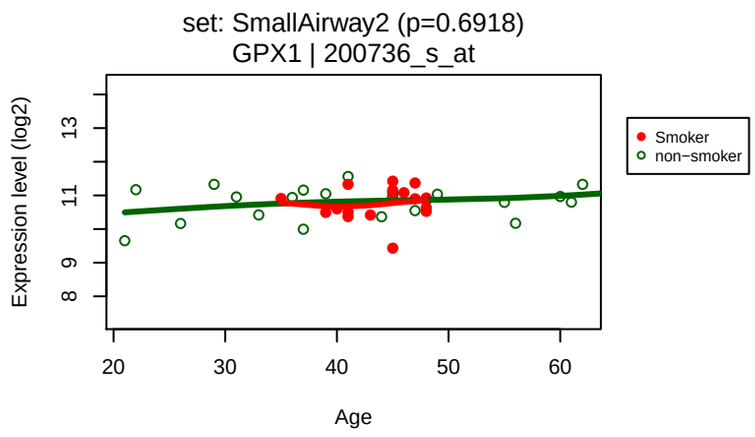
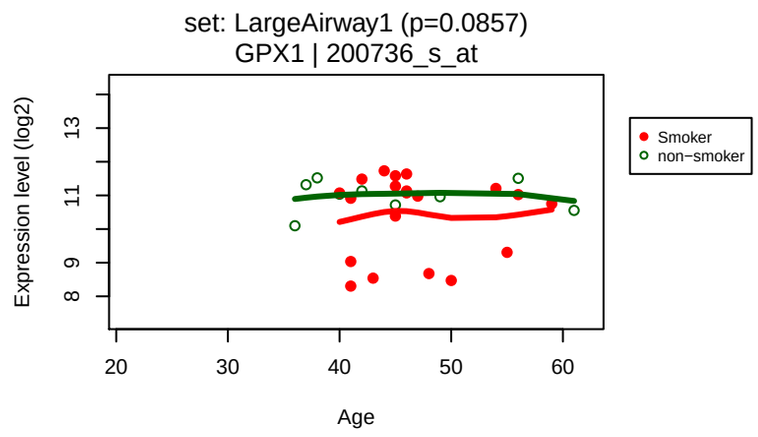

Gene expression of smokers and nonsmokers
gene: GPX1 | 200736_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4776
p=0.6918
p=0.9645
p=0.0857
p=0.8886
p=0.8028
p=0.2431
p=0.0783
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

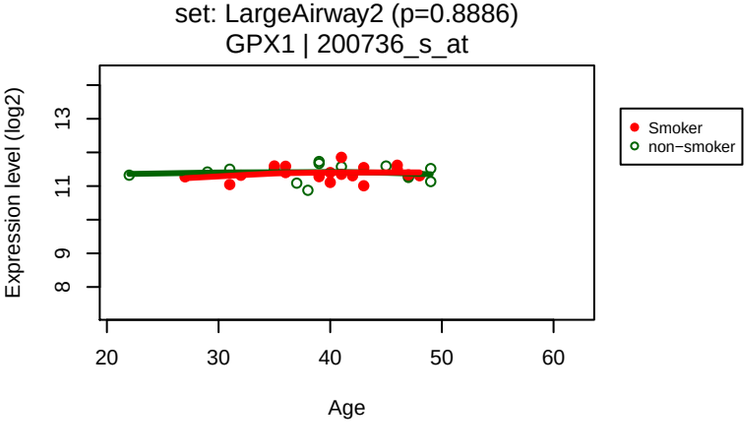

Gene expression ratio (smokers/nonsmokers)
gene: GPX1 | 200736_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4776
p=0.6918
p=0.9645
p=0.0857
p=0.8886
p=0.8028
p=0.2431
p=0.0783
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

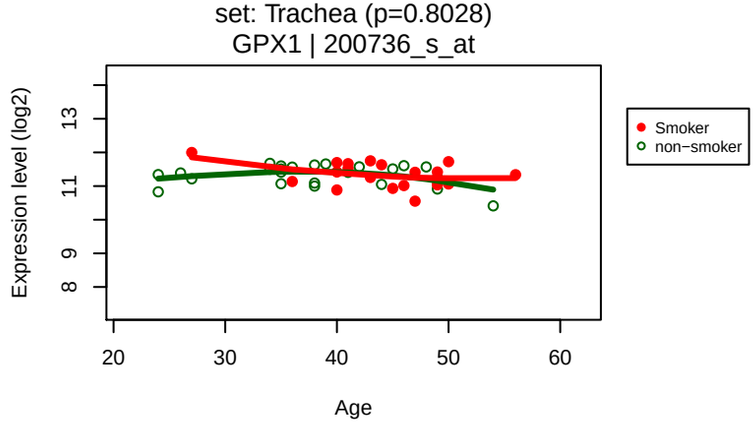

Gene expression ratio in Males
gene: GPX1 | 200736_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4675
p=0.5599
p=0.9956
p=0.031
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPX1 | 200736_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8887
p=0.202
p=0.4204
p=0.8044
SmallAirway2
LargeAirway2
Trachea
Macrophage2

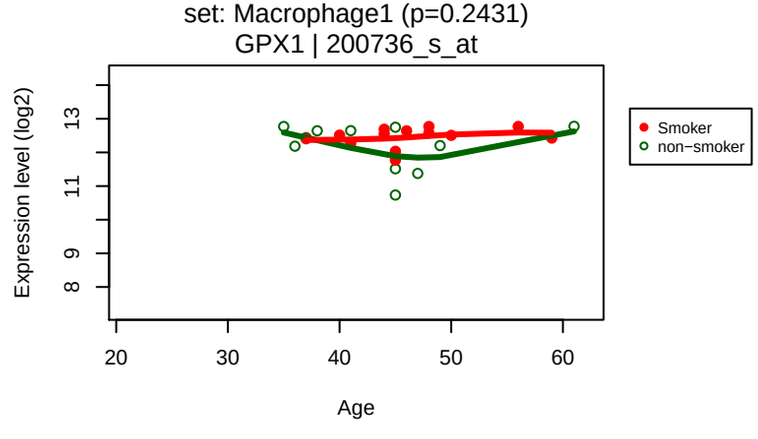

set: SmallAirway1 (p=0.4776)
GPX1 | 200736_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker

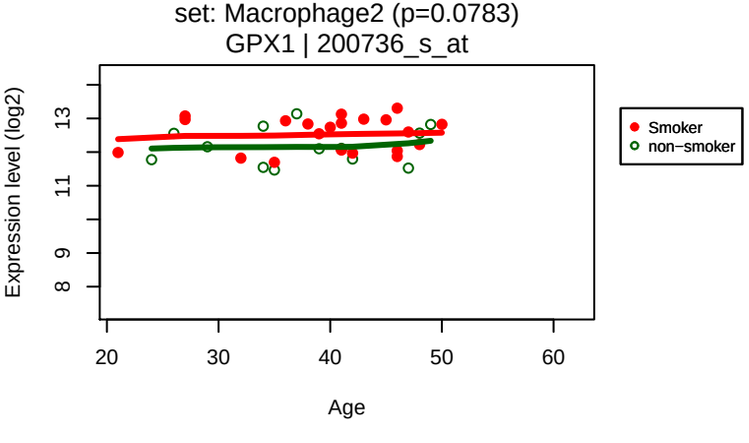

set: SmallAirway2 (p=0.6918)
GPX1 | 200736_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker

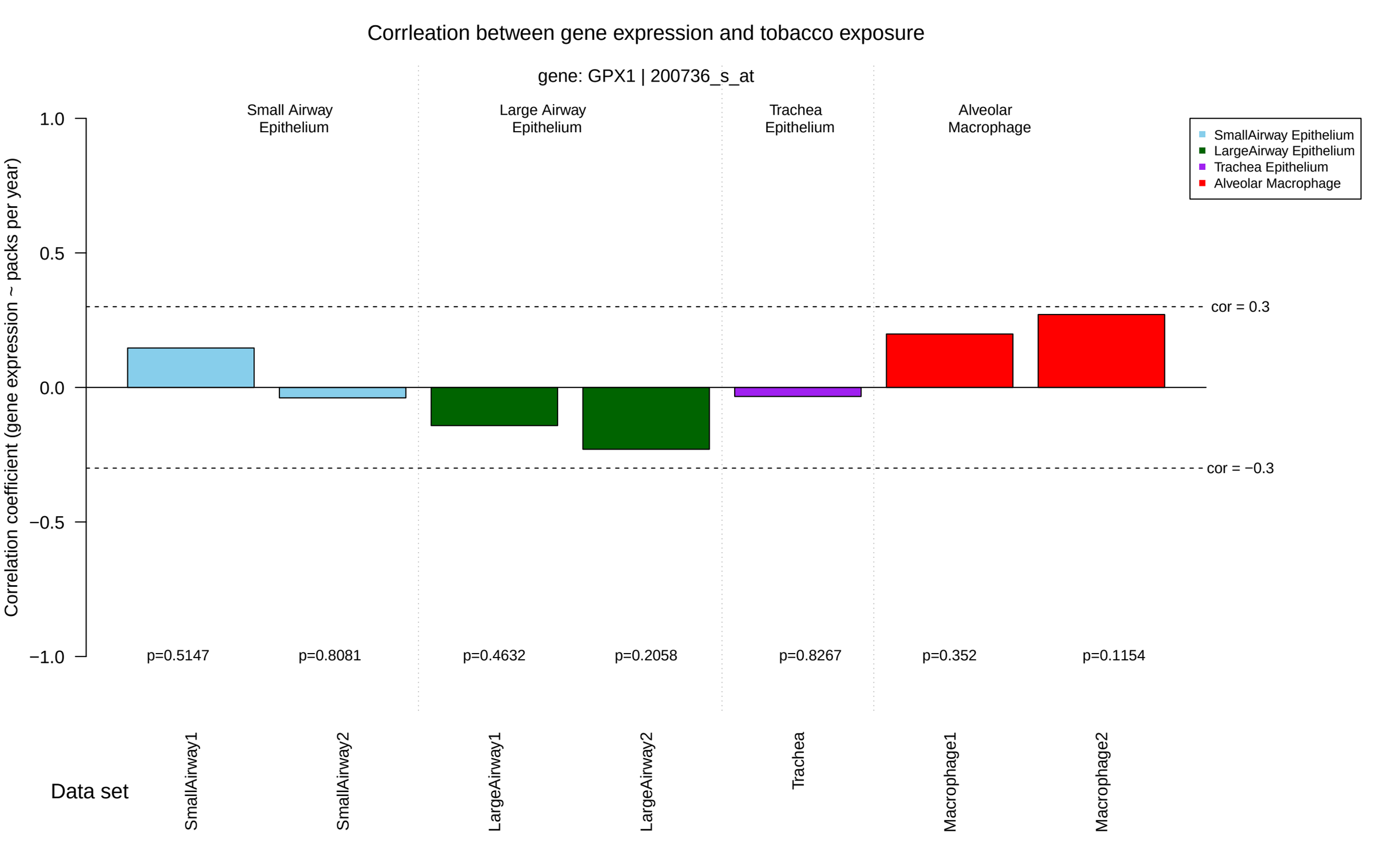

set: LargeAirway1 (p=0.0857)
GPX1 | 200736_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8886)
GPX1 | 200736_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.8028)
GPX1 | 200736_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2431)
GPX1 | 200736_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0783)
GPX1 | 200736_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPX1 | 200736_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5147
p=0.8081
p=0.4632
p=0.2058
p=0.8267
p=0.352
p=0.1154
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2